

HW5 – Design D0

Team Members:

Noah Schumacher

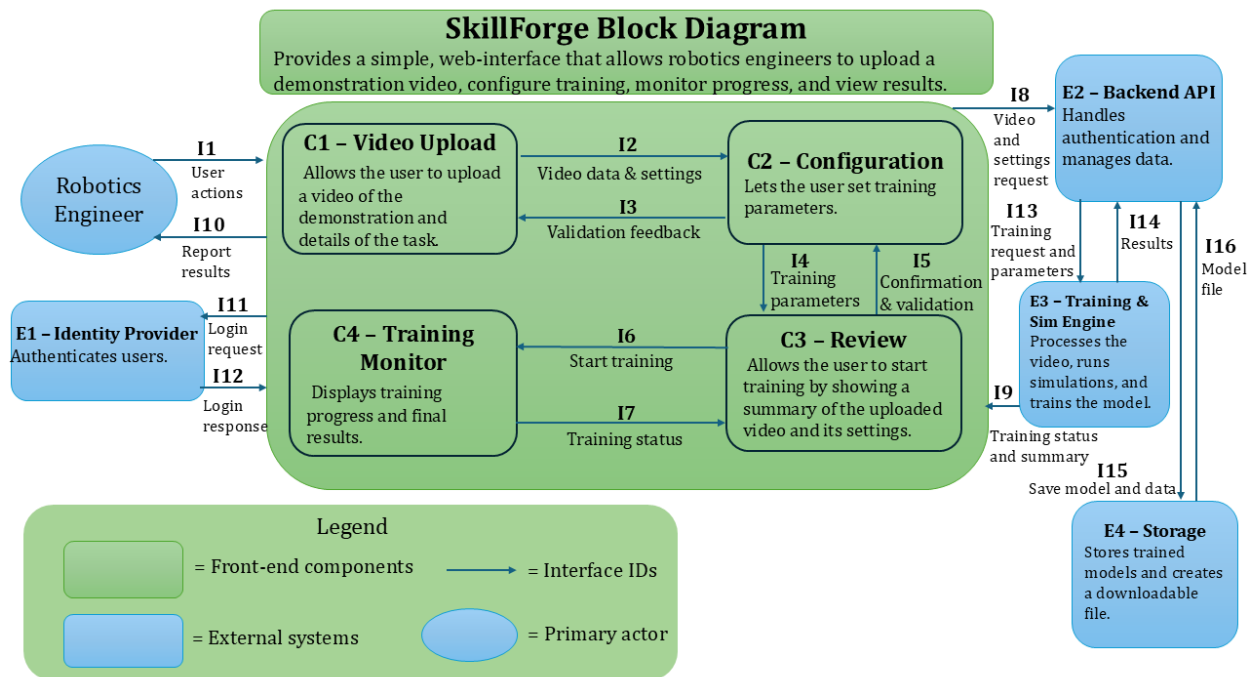
Ryan Broyles

Manjinder Kaur

Minh Nguyen

Akshat Chaturvedi

Block Diagram



Component Responsibility Table

Component Name	Component Purpose	Component Interfaces	Team Owner
E1 - Identity Provider	Allow for creation, management, and validation of auth tokens	Interfaces In: I11 Interfaces Out: I12	Manjinder Kaur
E2 - Backend API	Manage communication between ML pipeline and frontend	Interfaces In: I8, I14 Interfaces Out: I13	Akshat Chaturvedi
E3 - Training & Sim Engine	Processes video, runs simulations, and trains model	Interfaces In: I13 Interfaces Out: I14, I9	Noah Schumacher
E4 - Storage	Stores trained models and cached video/point cloud data	Interfaces In: I15 Interfaces Out: I16	Ryan Broyles
C1 – Video Upload	Frontend interface for uploading training data and visualizing upload process	Interfaces In: I1 Interfaces Out: I10	Manjinder Kaur
C2 - Configuration	Interface for allowing users to configure custom configuration settings	Interfaces In: I2, I5 Interfaces Out: I3, I4	Ryan Broyles
C3 - Review	Training config and data usage confirmation screen	Interfaces In: I7, I4 Interfaces Out: I6, I5	Minh Nguyen
C4 – Training Monitor	Visualize the training and sim process, show any training errors	Interfaces In: I6 Interfaces Out: I7	Minh Nguyen

Interface Specification Table

ID	Connection	Input	Output	Format	Protocol	Error handling
I1	Primary actor > C1	Video file, task details	Uploaded video and details	MP4/MOV, plain text	UI	Displays error if the video is

						not supported or is missing.
I2	C1 > C2	Video file, task details	Video info	MP4/MOV, plain text	Function call	Displays error if the video is not available.
I3	C2 > C1	Video info	Validation message	Plain text	Function call	Displays validation message for corrections.
I4	C2 > C3	Training settings	Reviewable settings	Plain text	Function call	Determines missing/invalid settings
I5	C3 > C2	Requested settings	Updated settings	Plain text	Function call	Settings will remain unchanged if failed.
I6	C3 > C4	Start training	Request confirmation	Plain text	Function call	Displays an error if it can't be started.
I7	C4 > C3	Training status	Status summary	Plain text	Function call	Displays the most recent status update.
I8	C2 > E2	Video and training settings	Request confirmation	MP4/MOV, plain text	API Call	An error is displayed and allows for a retry.
I9	E3 > C4	Training status and results	Status update	Plain text	API Call	Displays a message that the status is unavailable.
I10	C4 > Primary actor	Training status and results	Progress report	Plain text	UI	Error is displayed if the report is unavailable.
I11	Primary actor > E1	Login Credentials	Login Authentication	Plain Text	API Call	Displays login error and confirmation message to try again.
I12	E1 > Primary actor	Login Confirmation	Identification Details	Plain Text	API Call	Displays server error and allows for a reconnection to server.

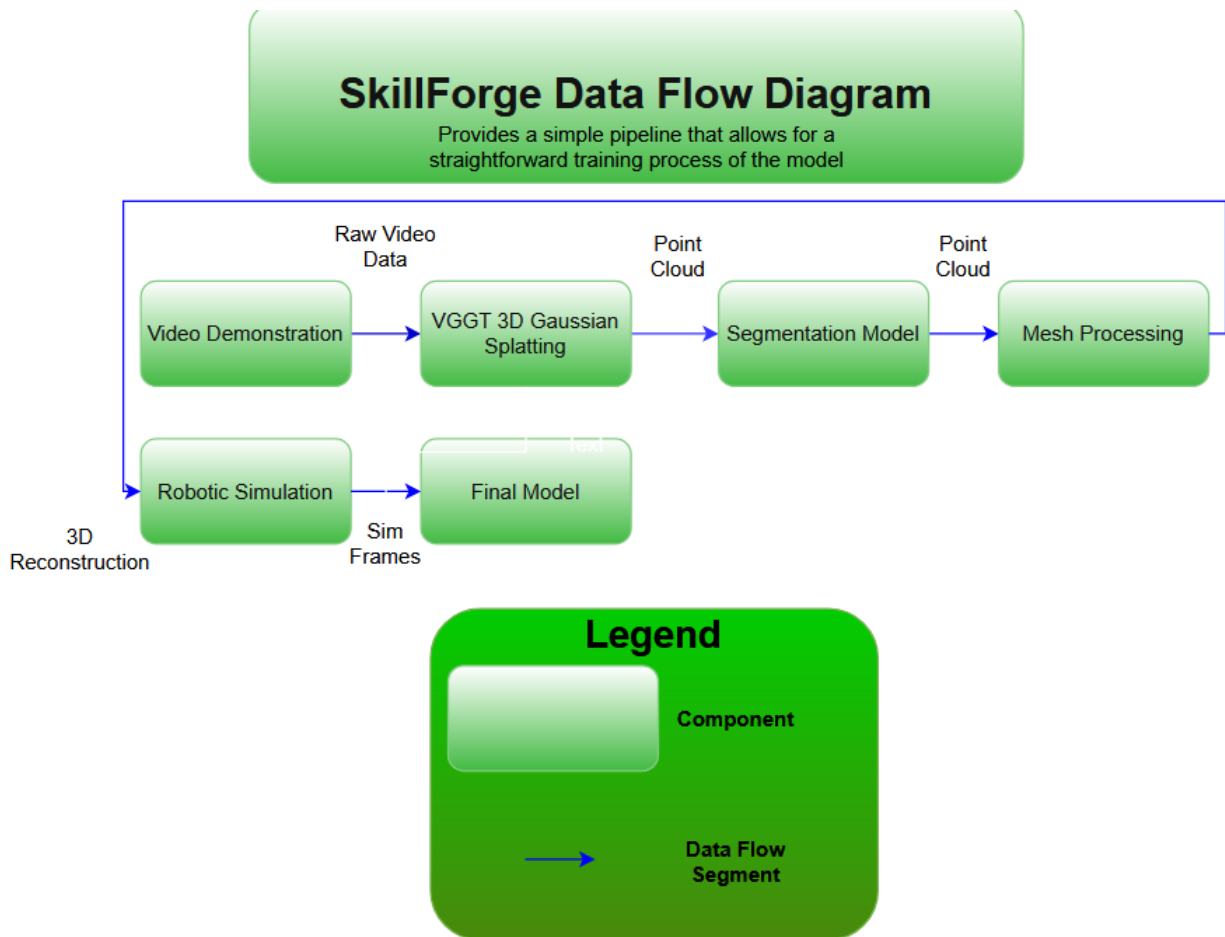
I13	E2 > E3	Request and Parameters for Training	Information about current training process	Plain Text	API Call	Displays information about failed training request for parameter correction.
I14	E3 > E2	Result information	Trained Model Specifications	Plain Text	API Call	Error message about failed training.
I15	E2 > E4	Trained Model Data	Confirmation of successful transfer	PT (PyTorch file)	FTPS	Displays failed transfer message and a request to try again
I16	E4 > E2	Model File	Validation of downloaded model	ONNX/PT	FTPS	Deletes previous downloadable model file and recreates a new one for fresh transfer.

Example payload:

I9 -

```
{
  "trainingID": "020",
  "status": "In Progress",
  "progressRate": "65%"
}
```

Data-flow Diagram



Architecture Pattern and Justification – Pipeline Architecture

Project SkillForge will use a pipeline architecture due to the linear flow of data through the system. In the architecture, video data is processed in a consistent manner across standard stages, with specified artifacts being saved to disk at each stage.

Criteria	Score Justification	Score (out of 10)
fit to the problem	Linear data processing path executed with limited interaction with frontend and	10/10

	frontend-facing backend components	
team skills	Team is equally proficient with all available pipelines	5/10
performance and timing	Pipeline arch reduces communication overhead from api calls and simplifies pipeline speed profiling	8/10
scalability	Pipeline arch allows for easy duplication of action across different compute resources, allowing for easy scaling to multiple simultaneous customer requests	9/10
hardware constraints	The primary hardware constraint will be the need to use cloud compute to run the ML pipeline. The pipeline architecture should allow for the ML pipeline to easily be containerized for cloud deployment.	7/10

Decision Log

Decision Name	Decision Description	Decision Alternative	Decision Rational
Architecture Type	Chose to use pipeline architecture	Use client-server architecture	Our system does not have enough frontend<->backend communication requirement to justify building the architecture around that requirement.
Video File Format	Support MP4 and MOV as primary video formats	AVI and WMV	Video files are the only non-text input to the system, and thus are the main data type we do not directly control. Supporting MP4 and MOV is chosen to allow for as much capability as we can. We chose

			not to support other formats to limit the complexity of the backend and storage components.
3D Reconstruction Method	Use VGGT->point cloud	Traditional gaussian splatting	VGGT offers a more out-of-the box methodology that is faster to run and tends to work better in dynamic scenes (scenes with motion, complex background)
Parameter Selection	Add component to allow users to configure ML pipeline parameters	Keep parameters hidden and only use default/system chosen parameters	Hyperparameter configuration was a core feature request seen in the user stories which is why it was included. Keeping parameters hidden is useful for protecting our pipelines design from being copied, but the use-case was given priority due to the strong desire for that functionality.